EMPLOYEE DETAILS AND PAYROLL MANAGEMENT SYSTEM

IT19341 – Introduction to OOP and JAVA project report

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RAJALAKSHMI ENGINEERING COLLEGE

BONAFIDE CERTIFICATE

Certified that this project report titled "**SPLITEASY**" is the bonafide work of "**DEBORAH JOAN SR (251001043), DIKSHA NAGARAJAN (251001057), DHIYA V(251001056)**" who carried out the project work under my supervision.

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This project report is submitted for IT19341 – Introduction to OOP and Java held on ……..
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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

SplitEasy is a smart expense-tracking and bill-splitting application intended to make shared spending easier to record, understand and settle. Friends, classmates, roommates and travelling groups often share costs but maintain the details through messages, notes or manual calculations. This can lead to missed expenses, incorrect shares and uncertainty about who owes whom.

The project combines an Android application developed using Java and XML with a backend service built using Spring Boot and REST APIs. MySQL is used for persistent data storage and is hosted on Aiven. Maven is used for backend build and dependency management. Docker configuration has been added, and the backend has been deployed on Render. Git and GitHub are used for source-code version control and collaboration.

The planned functional areas include account access and profile management, group creation and member management, personal and shared expense recording, equal/exact/percentage split options, balance calculation, settlement records and expense insights. Additional group-supporting workflows include reminders, events, guest records, calendar and polls, shared documents, shopping lists, chores and trip expenses.

Backend setup, database connectivity and deployment health checks have been established. The report distinguishes these verified setup milestones from features that still require complete Android-to-backend, security, notification and payment-flow testing. The objective is to provide a maintainable foundation for transparent shared-expense management.

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1.- INTRODUCTION

PROBLEM STATEMENT

Shared expenses are common in student groups, families, trips and shared accommodation. One person may pay for food, another may pay for transport, and a third may pay for tickets or supplies. When these transactions are recorded informally, participants may forget amounts, duplicate entries or disagree about how a total should be divided.

Manual calculations become more difficult as the number of members and transactions increases. Equal division is not always appropriate: some expenses may involve only selected members, some members may have agreed to exact amounts, and other costs may be divided by percentage. A useful system must preserve the payer, participants, amount, date, category and split details so that balances can be explained later.

SplitEasy addresses this problem by providing a common digital place to organize groups and expenses, calculate shares and record settlements. The application aims to reduce arithmetic effort and improve transparency. The system does not itself prove that an external payment has occurred unless a verified payment integration is implemented; a settlement record should therefore be treated as an application record until independently confirmed.

1.2 PROJECT OVERVIEW

SplitEasy is designed as an Android-based expense-sharing application supported by a Spring Boot REST backend and a MySQL database. Users can organize expenses around groups such as a class outing, a shared apartment or a trip. Each expense records who paid and how the cost is allocated. The resulting shares contribute to member balances, which can be reviewed and settled.

1.3 PURPOSE

The purpose is to replace scattered notes and repeated manual calculations with a structured record of shared spending. The system is intended to make group activity easier to coordinate while keeping expense information accessible to authorized members.

1.4 OBJECTIVES

- Provide account registration, login and profile-related workflows.
- Allow users to create groups and manage members.
- Record personal and shared expenses with relevant details.
- Support equal, exact-amount and percentage allocation where implemented.
- Calculate balances from expense shares and recorded settlements.
- Present expense history and category-based insights.
- Support group coordination through reminders, events, polls, documents and trip records.
- Maintain persistent records through the backend and MySQL database.

1.5 SCOPE

The project covers an Android client, a backend API layer and a relational database. The mobile application is the primary client in the current implementation scope. A React web client is a possible future extension and must not be described as completed unless a working implementation is available.

2. FUNCTIONALITIES PROVIDED

The functional scope is grouped into the following areas:

• Account and profile: registration, login, profile information and account/PIN recovery workflows.

• Groups: create a group, view group information and manage membership.

• Expenses: create personal or shared expense records, identify the payer, enter amount and category, and select participants.

• Splitting: allocate an expense equally, by exact amount or by percentage, subject to the implemented validation rules.

• Balances and settlements: summarize net amounts owed or receivable and store settlement entries.

• Insights: summarize spending and expense history by available categories and periods.

• Group coordination: reminders, announcements, events, guests, calendar entries, polls, documents, shopping lists and chores.

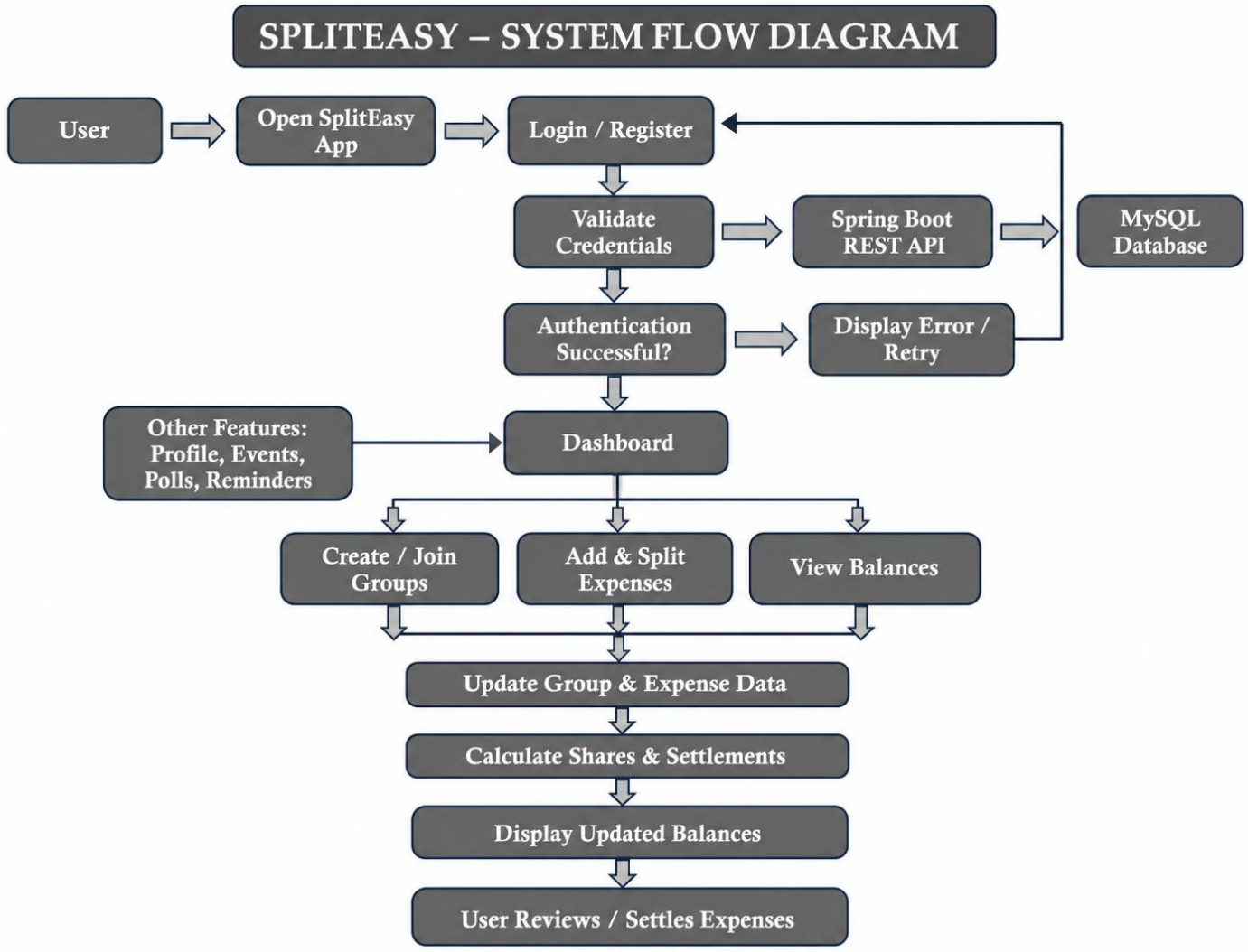
• Trips: maintain trip-related information and trip expenses.

3. - SYSTEM FLOW DIAGRAM

System flow diagram helps in understanding the flow of information and processes within the system. The below displays such a representation for the project on Split Easy expense-sharing management system.

3.1 DATAFLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual way to show how data moves through a system or process. The below displays such a representation for the project on Split Easy expense-sharing management system.

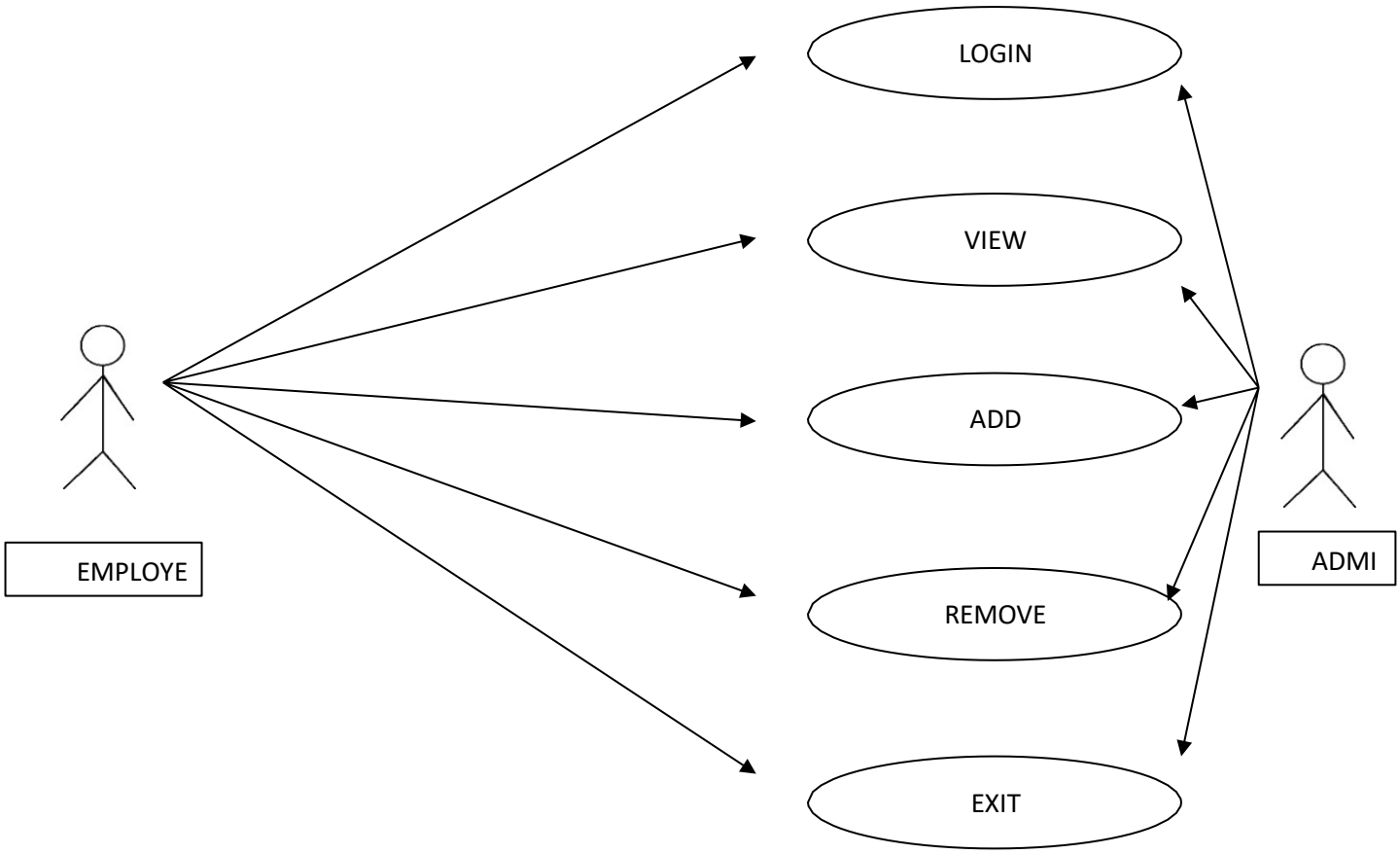


3.2 USE CASE DIAGRAM

A use case diagram is a visual representation that illustrates the interactions between actors (users) and the SplitEasy system. It depicts the various ways users interact with the application, highlighting the functionalities it provides, such as user registration and login, group creation and management, expense recording and splitting, balance calculation, settlement tracking, and viewing expense insights. The diagram provides an overview of how users utilize SplitEasy to manage shared expenses and simplify bill-splitting activities within a group.

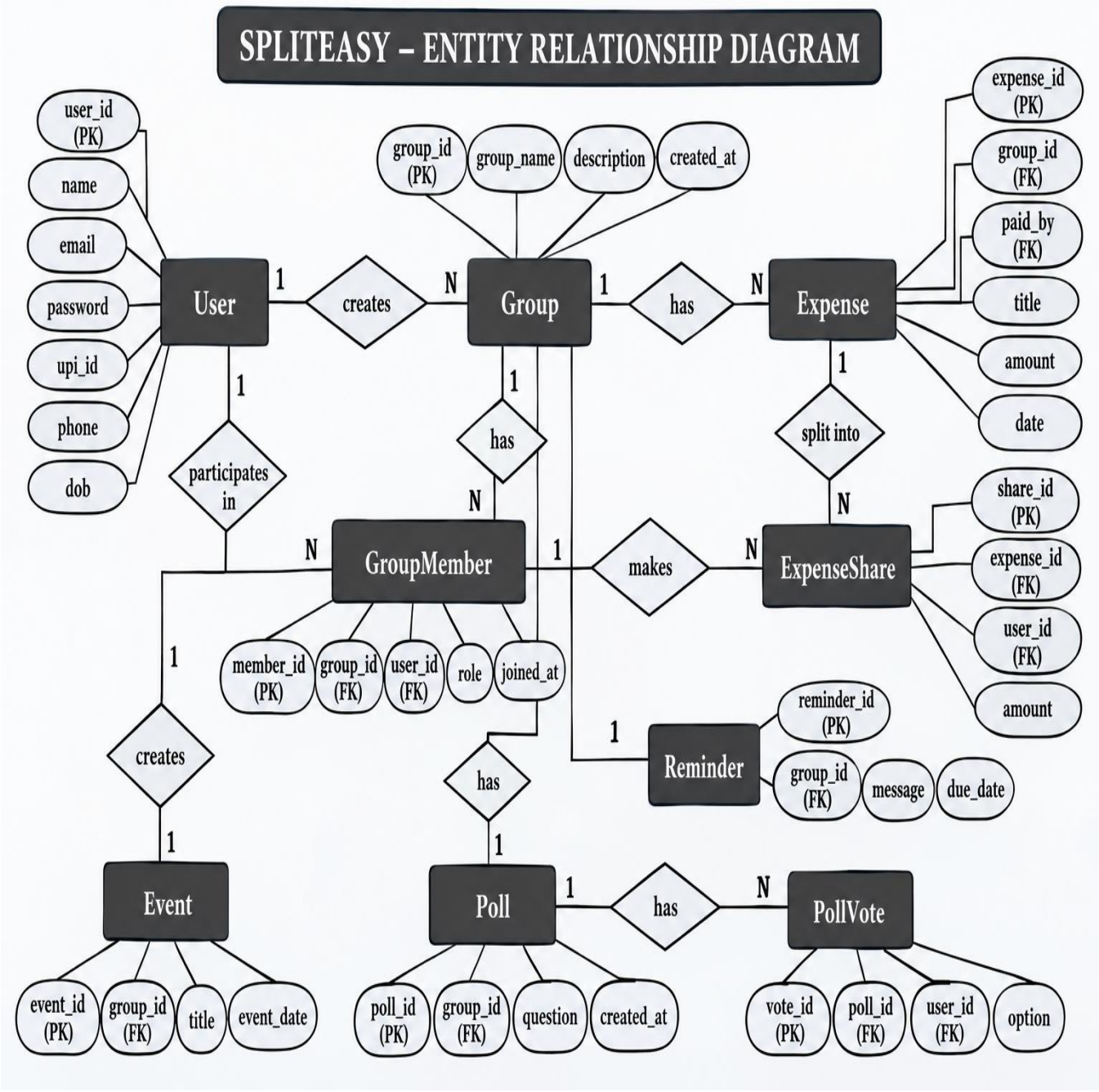
A

ctor:
Employee
Use Cases
Add Employee - Description: Allows an employee to add new employee details to the system.
Remove Employee - Description: Permits an employee to remove an existing employee from the system.
View Employee Details - Description: Enables an employee to view details of other employees.
Exit System - Description: Allows an employee to log out or exit the system.



3.3 ENTITY RELATIONSHIP DIAGRAM

An Entity-Relationship Diagram (ERD) is a visual representation that illustrates the relationships between different entities within a database system. The below diagram displays the ERD for the Split Easy expense-sharing management system.



4. - MODULE DESCRIPTION

The different types of modules present in the SplitEasy project are:

SPLASH PAGE:
Displays the SplitEasy application name and logo when the application starts. It acts as the introductory screen before users navigate to login or registration.

USER REGISTRATION AND LOGIN:
Allows new users to register by providing the required details and existing users to log in. It provides account access to the application's features.

GROUP MANAGEMENT:
Allows users to create groups for friends, family or other shared activities. Users can view and manage their group information.

GROUP MEMBER MANAGEMENT:
Allows users to manage members associated with each group. It helps identify participants involved in shared expenses and group activities.

ADD EXPENSE:
Allows users to record personal or shared expenses by entering details such as description, amount, date and payer. Users can select the group members involved in an expense.

EXPENSE SPLITTING:
Supports equal, exact-amount and percentage-based expense splitting. It helps distribute the total expense among selected members according to the chosen method.

VIEW EXPENSES:
Allows users to view recorded personal and group expenses. It helps members track spending and review individual expense details.

BALANCE CALCULATION:
Calculates how much each member owes or is owed based on recorded expenses and shares. It provides a clear summary of outstanding group balances.

SETTLEMENT:
Allows users to view pending balances and record repayments between group members. It helps track settlement records and remaining amounts.

INSIGHTS AND EXPENSE REPORTS:
Provides summaries of expenses and available spending information. It helps users understand spending patterns and review expense distribution.

PAYMENT REMINDERS:
Helps users maintain reminders for pending payments and outstanding balances. It supports better follow-up and tracking of group payments.

EVENT AND GROUP MANAGEMENT:
Allows users to organize group events and maintain related guest information. It helps members coordinate activities and manage event details.

POLLS AND ANNOUNCEMENTS:
Supports polls for group decisions and announcements for sharing important information. It improves communication and coordination among group members.

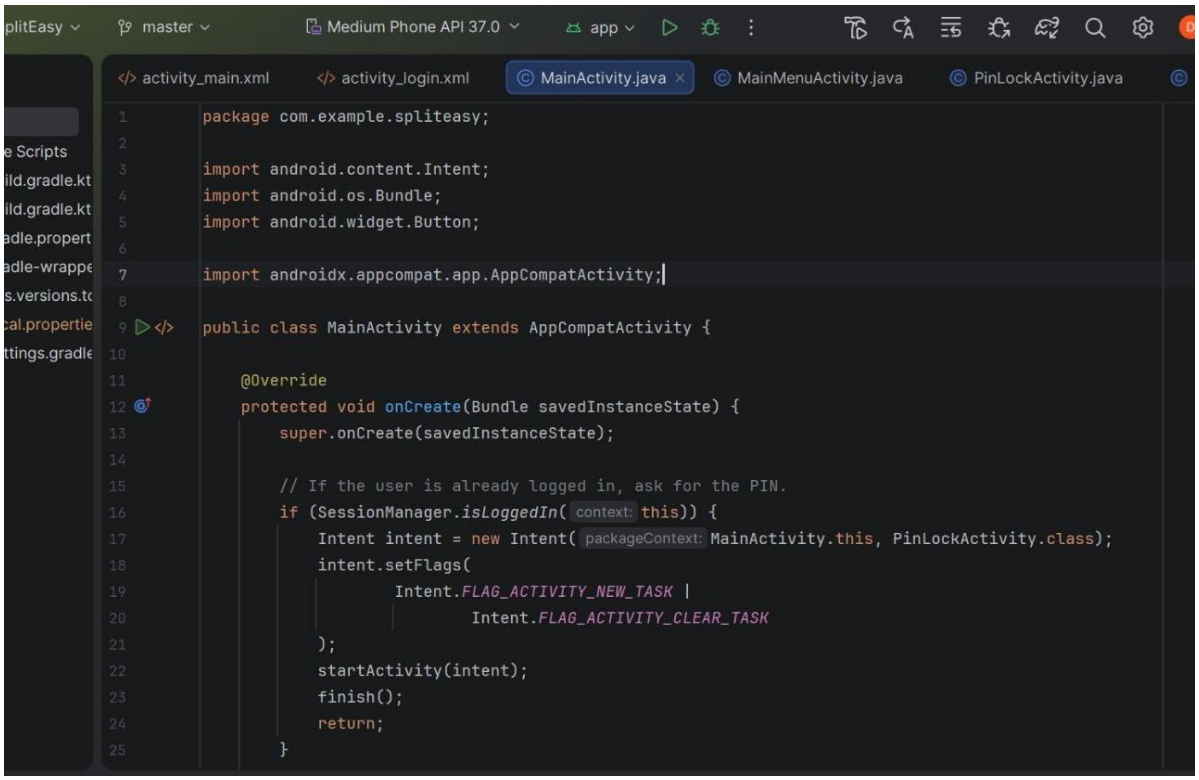
GROUP DOCUMENTS, SHOPPING LISTS AND CHORES:
Provides support for organizing shared documents, shopping lists and group responsibilities. It helps members manage common resources and tasks.

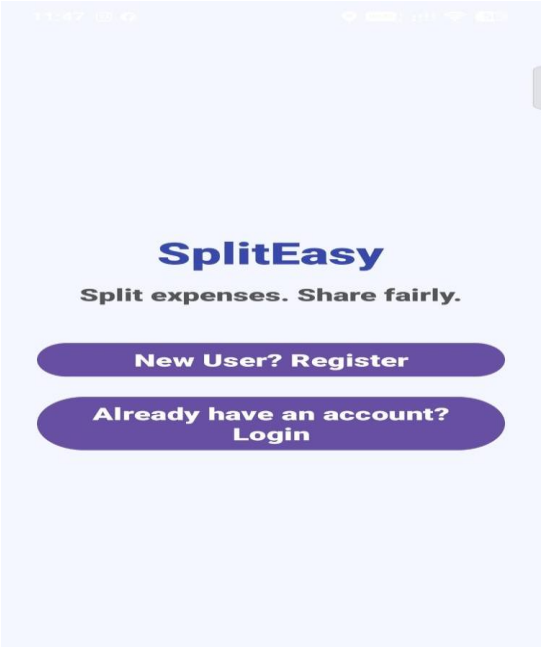
TRIP MANAGEMENT:
Allows users to organize trip-related information and maintain trip expenses. It helps groups track spending during shared travel activities.

BACKEND AND DATABASE MANAGEMENT:
Uses Spring Boot REST APIs for backend operations and MySQL hosted on Aiven for data storage. Maven, Docker, Render and GitHub support building, deployment and version control.

- IMPLEMENTATION

SPLASH PAGE/ MAIN ACTIVITY :





JAVA CODE:

```
package com.example.spliteasy;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

import
androidx.appcompat.app.AppCompat
Activity;

public class MainActivity extends
AppCompatActivity {

   @Override
   protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);

       // If the user is already logged in,
ask for the PIN.
       if
(SessionManager.isLoggedIn(this)) {
           Intent intent = new
Intent(MainActivity.this,
PinLockActivity.class);
           intent.setFlags(

Intent.FLAG_ACTIVITY_NEW_TAS
K |

Intent.FLAG_ACTIVITY_CLEAR_T
ASK
           );
           startActivity(intent);
           finish();
           return;
       }

       // Otherwise, show the login and
registration screen.

setContentView(R.layout.activity_mai
n);

       Button registerButton =
findViewById(R.id.registerButton);
       Button loginButton =
findViewById(R.id.loginButton);
```

```
        registerButton.setOnClickListener(v -
        > {
                startActivity(new
        Intent(MainActivity.this,
        RegisterActivity.class));
            });

            loginButton.setOnClickListener(v
        -> {
                startActivity(new
        Intent(MainActivity.this,
        LoginActivity.class));
            });
        }
        }
```

- **ANDROID MAIN ACTIVITY:**

```
package com.example.spliteasy;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Check whether the user is already logged in
        if (SessionManager.isLoggedIn(this)) {

            Intent intent = new Intent(
                    MainActivity.this,
                    PinLockActivity.class
            );

            intent.setFlags(
                    Intent.FLAG_ACTIVITY_NEW_TASK |
                    Intent.FLAG_ACTIVITY_CLEAR_TASK
            );

            startActivity(intent);
            finish();
```

```
            return;
        }

        // Display the main welcome screen
        setContentView(R.layout.activity_main);

        Button registerButton = findViewById(R.id.registerButton);
        Button loginButton = findViewById(R.id.loginButton);

        registerButton.setOnClickListener(v -> {
            startActivity(new Intent(
                    MainActivity.this,
                    RegisterActivity.class
            ));
        });

        loginButton.setOnClickListener(v -> {
            startActivity(new Intent(
                    MainActivity.this,
                    LoginActivity.class
            ));
        });
    }
}
```

SPRING BOOT MAIN CLASS :

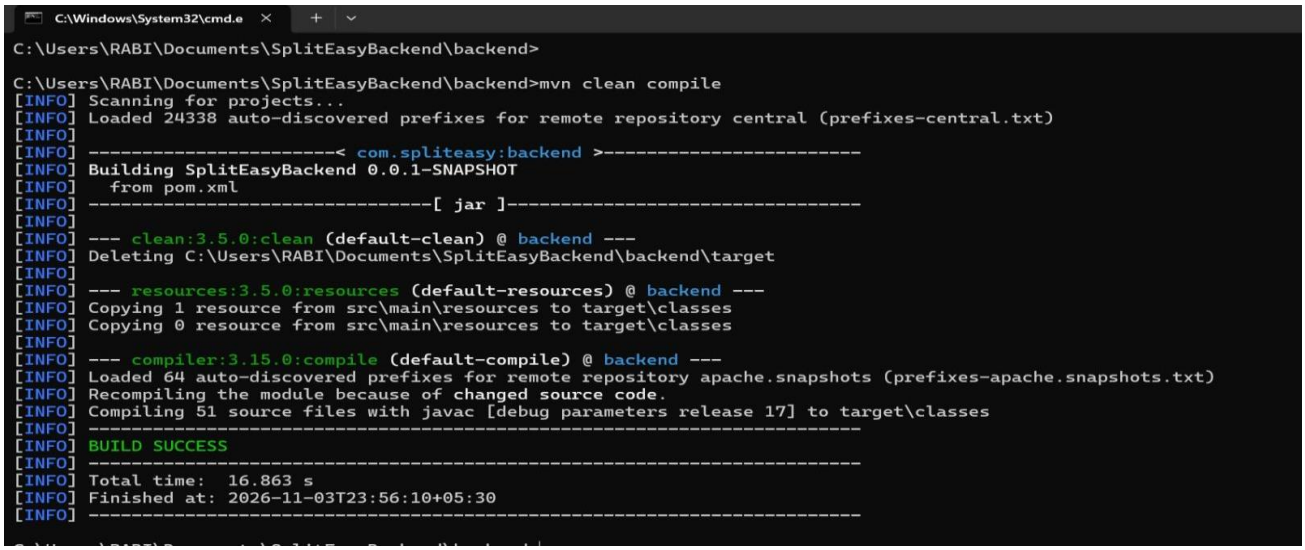
```
package com.spliteasy.backend;

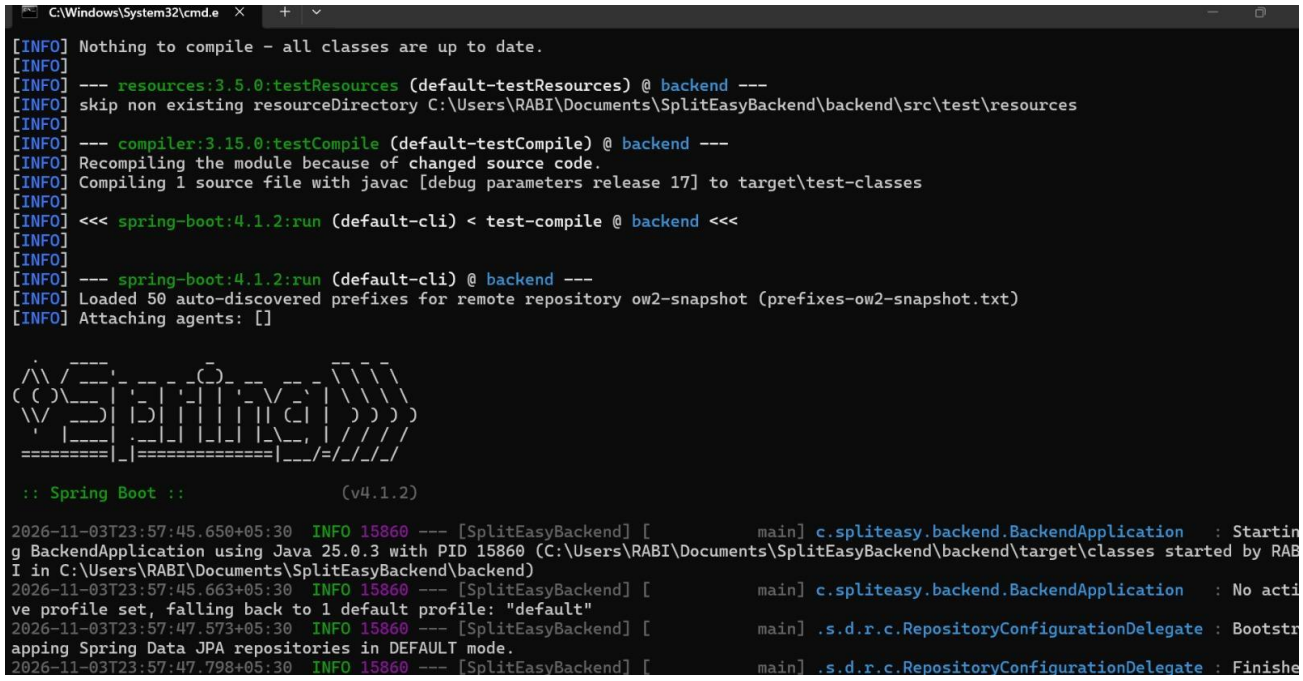
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class BackendApplication {

    public static void main(String[] args) {
        SpringApplication.run(BackendApplication.class, args);
    }

}
```





JDBC CONNECTION :

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {

    private static final String URL =
            "jdbc:mysql://localhost:3306/spliteasy";

    private static final String USER = "root";
    private static final String PASSWORD = "YOUR_PASSWORD";

    public static Connection getConnection() {

        Connection con = null;

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

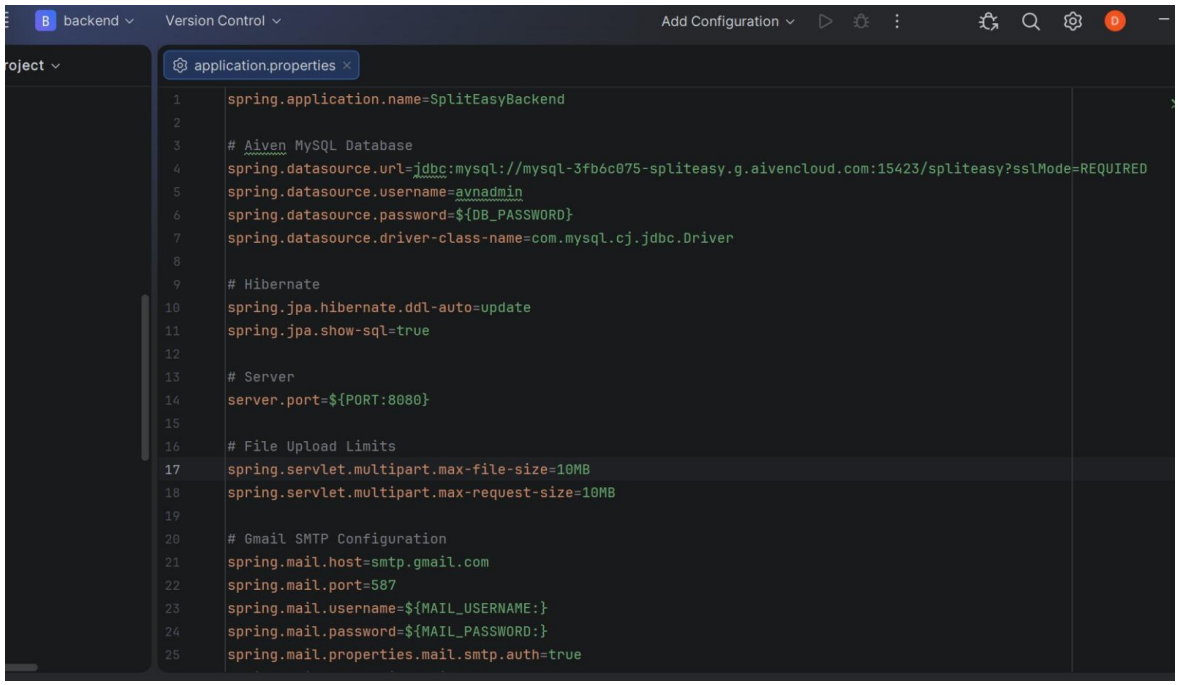
            con = DriverManager.getConnection(
                    URL, USER, PASSWORD
            );

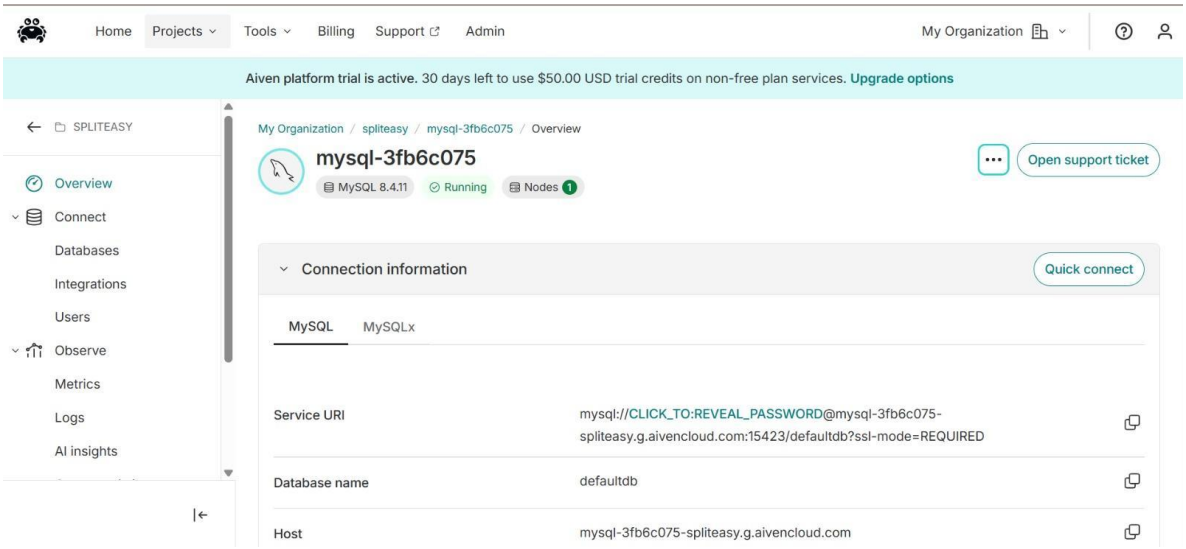
            System.out.println("Database Connected Successfully!");

        } catch (ClassNotFoundException | SQLException e) {
            System.out.println("Database Connection Failed!");
            e.printStackTrace();
        }

        return con;
    }

    public static void main(String[] args) {
        getConnection();
    }
}
```


backend
Version Control
Add Configuration
application.properties
spring.application.name=SplitEasyBackend
Aiven MySQL Database
spring.datasource.url=jdbc:mysql://mysql-3fb6c075-spliteasy.g.aivencloud.com:15423/spliteasy?sslMode=REQUIRED
spring.datasource.username=avnadmin
spring.datasource.password=${DB_PASSWORD}
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
Hibernate
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
Server
server.port=${PORT:8080}
File Upload Limits
spring.servlet.multipart.max-file-size=10MB
spring.servlet.multipart.max-request-size=10MB
Gmail SMTP Configuration
spring.mail.host=smtp.gmail.com
spring.mail.port=587
spring.mail.username=${MAIL_USERNAME:}
spring.mail.password=${MAIL_PASSWORD:}
spring.mail.properties.mail.smtp.auth=true


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mysql-3fb6c075
MySQL 8.4.11
Running
Nodes 1
Open support ticket
Connection information
Quick connect
MySQL
MySQLx
Service URI
mysql://CLICK_TO:REVEAL_PASSWORD@mysql-3fb6c075-spliteasy.g.aivencloud.com:15423/defaultdb?ssl-mode=REQUIRED
Database name
defaultdb
Host
mysql-3fb6c075-spliteasy.g.aivencloud.com

LOGIN PAGE :

```
package com.example.spliteasy;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    EditText emailEditText, passwordEditText;
    Button loginButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        emailEditText = findViewById(R.id.emailEditText);
        passwordEditText = findViewById(R.id.passwordEditText);
        loginButton = findViewById(R.id.loginButton);

        loginButton.setOnClickListener(v -> {

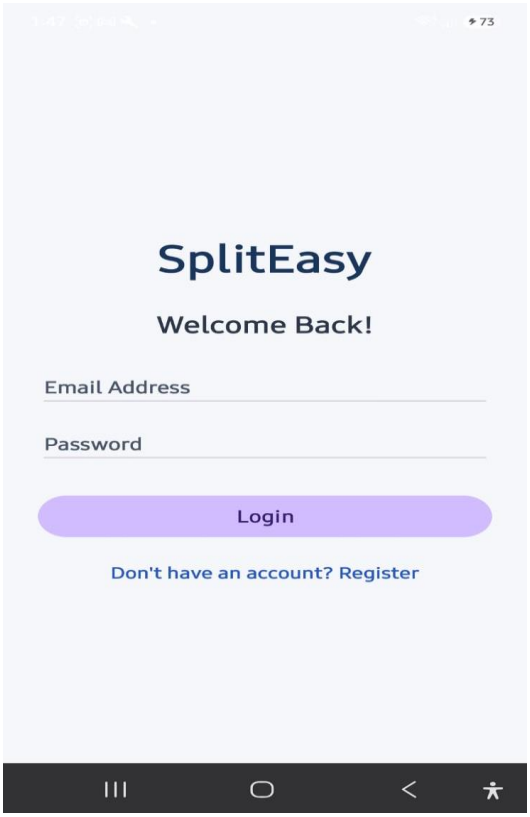
            String email = emailEditText.getText().toString().trim();
            String password = passwordEditText.getText().toString().trim();

            if (email.isEmpty()) {
                emailEditText.setError("Enter your email");
                return;
            }

            if (password.isEmpty()) {
                passwordEditText.setError("Enter your password");
                return;
            }

            Toast.makeText(
                    LoginActivity.this,
                    "Please wait. Logging in...",
                    Toast.LENGTH_SHORT
            ).show();

            // Connect to the backend authentication API here.
        });
    }
}
```



GROUP CREATION :

```
package com.example.spliteasy;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class CreateGroupActivity extends AppCompatActivity {

    EditText groupNameEditText, descriptionEditText;
    Button createGroupButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_create_group);

        groupNameEditText = findViewById(R.id.groupNameEditText);
        descriptionEditText = findViewById(R.id.descriptionEditText);
        createGroupButton = findViewById(R.id.createGroupButton);

        createGroupButton.setOnClickListener(v -> {
```

```
        String groupName =
                groupNameEditText.getText().toString().trim();

        String description =
                descriptionEditText.getText().toString().trim();

        if (groupName.isEmpty()) {
            groupNameEditText.setError("Enter group name");
            return;
        }

        if (description.isEmpty()) {
            descriptionEditText.setError("Enter group description");
            return;
        }

        // Send group details to the backend API here.

        Toast.makeText(
                CreateGroupActivity.this,
                "Group details entered successfully",
                Toast.LENGTH_SHORT
        ).show();
      });
   }
}
```

GROUP MEMBER MANAGEMENT :

```
package com.example.spliteasy;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

public class GroupMembersActivity extends AppCompatActivity {

    EditText memberNameEditText;
    Button addMemberButton;
    ListView membersListView;

    ArrayList<String> members;
    ArrayAdapter<String> adapter;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
   super.onCreate(savedInstanceState);
   setContentView(R.layout.activity_group_members);

   memberNameEditText = findViewById(R.id.memberNameEditText);
   addMemberButton = findViewById(R.id.addMemberButton);
   membersListView = findViewById(R.id.membersListView);

   members = new ArrayList<>();

   adapter = new ArrayAdapter<>(
         this,
         android.R.layout.simple_list_item_1,
         members
   );

   membersListView.setAdapter(adapter);

   addMemberButton.setOnClickListener(v -> {

      String memberName =
            memberNameEditText.getText().toString().trim();

      if (memberName.isEmpty()) {
         memberNameEditText.setError("Enter member name");
         return;
      }

      members.add(memberName);
      adapter.notifyDataSetChanged();

      memberNameEditText.setText("");

      Toast.makeText(
            GroupMembersActivity.this,
            "Member added successfully",
            Toast.LENGTH_SHORT
      ).show();
   });

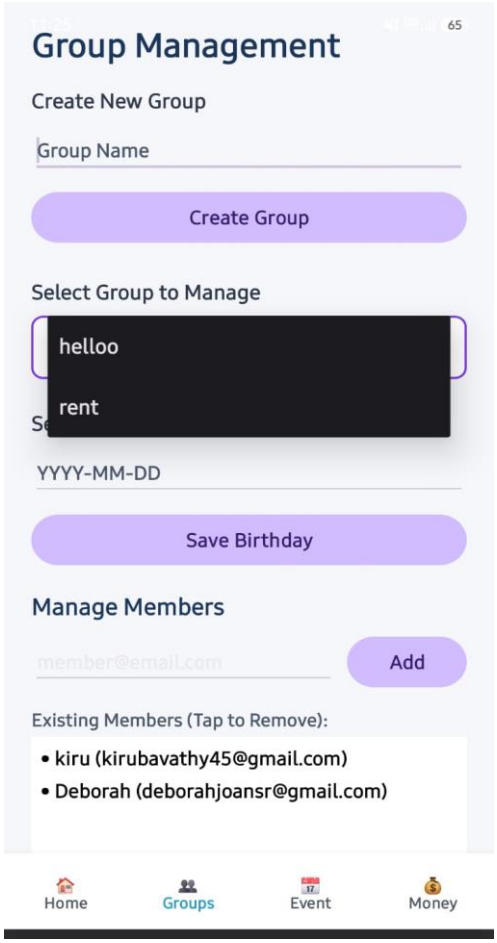
   membersListView.setOnItemLongClickListener(
         (parent, view, position, id) -> {

            members.remove(position);
            adapter.notifyDataSetChanged();

            Toast.makeText(
                  GroupMembersActivity.this,
                  "Member removed successfully",
                  Toast.LENGTH_SHORT
```

```
                ).show();

                return true;
            }
        );
    }
}
```



ADD EXPENSE :

```
package com.example.spliteasy;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class AddExpenseActivity extends AppCompatActivity {
```

```
EditText expenseNameEditText;
EditText amountEditText;
EditText paidByEditText;
Button addExpenseButton;

@Override
protected void onCreate(Bundle savedInstanceState) {
   super.onCreate(savedInstanceState);
   setContentView(R.layout.activity_add_expense);

   expenseNameEditText = findViewById(R.id.expenseNameEditText);
   amountEditText = findViewById(R.id.amountEditText);
   paidByEditText = findViewById(R.id.paidByEditText);
   addExpenseButton = findViewById(R.id.addExpenseButton);

   addExpenseButton.setOnClickListener(v -> {

      String expenseName =
            expenseNameEditText.getText().toString().trim();

      String amountText =
            amountEditText.getText().toString().trim();

      String paidBy =
            paidByEditText.getText().toString().trim();

      if (expenseName.isEmpty()) {
         expenseNameEditText.setError("Enter expense name");
         return;
      }

      if (amountText.isEmpty()) {
         amountEditText.setError("Enter amount");
         return;
      }

      if (paidBy.isEmpty()) {
         paidByEditText.setError("Enter payer name");
         return;
      }

      try {
         double amount = Double.parseDouble(amountText);

         if (amount <= 0) {
            amountEditText.setError("Enter a valid amount");
            return;
         }

         Toast.makeText(
               AddExpenseActivity.this,
               "Expense added successfully",
```

```
                Toast.LENGTH_SHORT
            ).show();

        } catch (NumberFormatException e) {
            amountEditText.setError("Enter a valid amount");
        }
    });
  }
}
```

VIEW EXPENSE :

```
package com.example.spliteasy;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

public class ViewExpenseActivity extends AppCompatActivity {

    ListView expenseListView;
    ArrayList<String> expenseList;
    ArrayAdapter<String> adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_view_expense);

        expenseListView = findViewById(R.id.expenseListView);

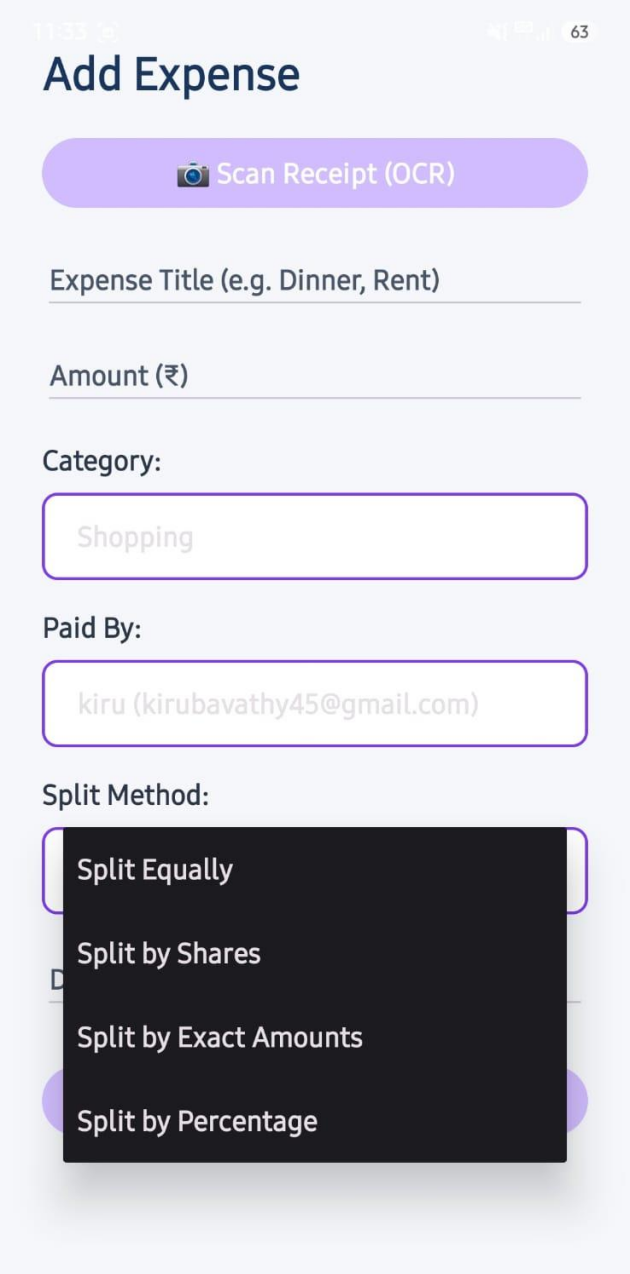
        expenseList = new ArrayList<>();

        // Sample expense details
        expenseList.add("Food - ₹500 - Paid by Rahul");
        expenseList.add("Travel - ₹300 - Paid by Priya");
        expenseList.add("Shopping - ₹700 - Paid by Arun");

        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                expenseList
        );

        expenseListView.setAdapter(adapter);
```

```
        if (expenseList.isEmpty()) {
            Toast.makeText(
                    this,
                    "No expenses available",
                    Toast.LENGTH_SHORT
            ).show();
        }
    }
}
```



BALANCE :

```
package com.example.spliteasy;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class SplitExpenseActivity extends AppCompatActivity {

    EditText totalAmountEditText, membersCountEditText;
    Button splitButton;
    TextView resultTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_split_expense);

        totalAmountEditText = findViewById(R.id.totalAmountEditText);
        membersCountEditText = findViewById(R.id.membersCountEditText);
        splitButton = findViewById(R.id.splitButton);
        resultTextView = findViewById(R.id.resultTextView);

        splitButton.setOnClickListener(v -> {

            String amountText =
                    totalAmountEditText.getText().toString().trim();

            String membersText =
                    membersCountEditText.getText().toString().trim();

            if (amountText.isEmpty()) {
                totalAmountEditText.setError("Enter total amount");
                return;
            }

            if (membersText.isEmpty()) {
                membersCountEditText.setError("Enter number of members");
                return;
            }

            try {
                double totalAmount = Double.parseDouble(amountText);
                int members = Integer.parseInt(membersText);

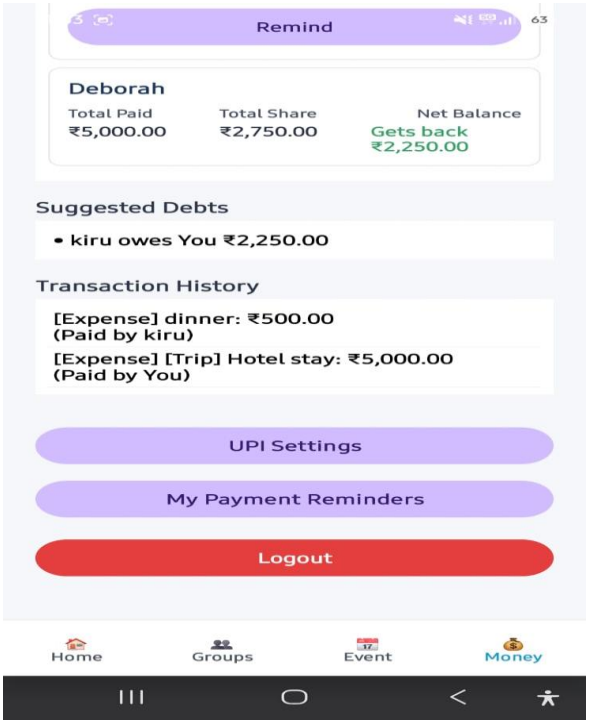
                if (totalAmount <= 0 || members <= 0) {
                    Toast.makeText(
```

```
                this,
                "Enter valid amount and member count",
                Toast.LENGTH_SHORT
            ).show();
            return;
        }

        double share = totalAmount / members;

        resultTextView.setText(
                "Total Expense: ₹" + totalAmount +
                "\nNumber of Members: " + members +
                "\nEach Member Should Pay: ₹" +
                String.format("%.2f", share)
        );

    } catch (NumberFormatException e) {
        Toast.makeText(
                this,
                "Please enter valid numbers",
                Toast.LENGTH_SHORT
        ).show();
    }
  });
 }
}
```



SETTLEMENT :

```
package com.example.spliteasy;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class SettlementActivity extends AppCompatActivity {

    EditText totalAmountEditText;
    EditText paidAmountEditText;
    EditText membersCountEditText;

    Button calculateButton;
    TextView settlementResultTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_settlement);

        totalAmountEditText = findViewById(R.id.totalAmountEditText);
        paidAmountEditText = findViewById(R.id.paidAmountEditText);
        membersCountEditText = findViewById(R.id.membersCountEditText);

        calculateButton = findViewById(R.id.calculateButton);
        settlementResultTextView =
                findViewById(R.id.settlementResultTextView);

        calculateButton.setOnClickListener(v -> {

            String totalText =
                    totalAmountEditText.getText().toString().trim();

            String paidText =
                    paidAmountEditText.getText().toString().trim();

            String membersText =
                    membersCountEditText.getText().toString().trim();

            if (totalText.isEmpty() || paidText.isEmpty()
                    || membersText.isEmpty()) {
                Toast.makeText(
                        this,
                        "Please fill all fields",
```

```
            Toast.LENGTH_SHORT
        ).show();
        return;
    }

    try {
        double total = Double.parseDouble(totalText);
        double paid = Double.parseDouble(paidText);
        int members = Integer.parseInt(membersText);

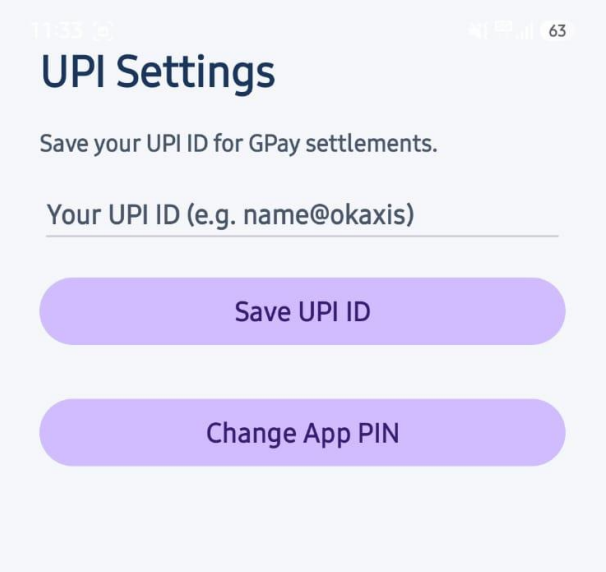
        if (total <= 0 || paid < 0 || members <= 0
                || paid > total) {
            Toast.makeText(
                    this,
                    "Enter valid settlement details",
                    Toast.LENGTH_SHORT
            ).show();
            return;
        }

        double share = total / members;
        double balance = paid - share;

        if (balance > 0) {
            settlementResultTextView.setText(
                    "Your share: ₹" +
                    String.format("%.2f", share) +
                    "\nYou should receive: ₹" +
                    String.format("%.2f", balance)
            );
        } else if (balance < 0) {
            settlementResultTextView.setText(
                    "Your share: ₹" +
                    String.format("%.2f", share) +
                    "\nYou should pay: ₹" +
                    String.format("%.2f", -balance)
            );
        } else {
            settlementResultTextView.setText(
                    "Your share: ₹" +
                    String.format("%.2f", share) +
                    "\nYour settlement is complete."
            );
        }

    } catch (NumberFormatException e) {
        Toast.makeText(
                this,
                "Please enter valid numbers",
                Toast.LENGTH_SHORT
        ).show();
    }
```

```
        });
    }
}
```



REST API CONTROLLER :

```
package com.spliteasy.backend.controller;

import com.spliteasy.backend.model.Expense;
import com.spliteasy.backend.repository.ExpenseRepository;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

import java.util.List;
import java.util.Optional;

@RestController
@RequestMapping("/api/expenses")
@CrossOrigin(origins = "*")
public class ExpenseController {

    @Autowired
    private ExpenseRepository expenseRepository;

    // Add a new expense
    @PostMapping
    public ResponseEntity<Expense> addExpense(
            @RequestBody Expense expense) {

        Expense savedExpense =
                expenseRepository.save(expense);
```

```
    return new ResponseEntity<>(
            savedExpense, HttpStatus.CREATED);
}

// Get all expenses
@GetMapping
public List<Expense> getAllExpenses() {

    return expenseRepository.findAll();
}

// Get expense by ID
@GetMapping("/{id}")
public ResponseEntity<Expense> getExpenseById(
        @PathVariable Long id) {

    Optional<Expense> expense =
            expenseRepository.findById(id);

    return expense.map(ResponseEntity::ok)
            .orElseGet(() ->
                    ResponseEntity.notFound().build());
}

// Update an expense
@PutMapping("/{id}")
public ResponseEntity<Expense> updateExpense(
        @PathVariable Long id,
        @RequestBody Expense updatedExpense) {

    Optional<Expense> existingExpense =
            expenseRepository.findById(id);

    if (existingExpense.isPresent()) {
        updatedExpense.setId(id);

        Expense savedExpense =
                expenseRepository.save(updatedExpense);

        return ResponseEntity.ok(savedExpense);
    }

    return ResponseEntity.notFound().build();
}

// Delete an expense
@DeleteMapping("/{id}")
public ResponseEntity<Void> deleteExpense(
        @PathVariable Long id) {

    if (expenseRepository.existsById(id)) {
```

```
            expenseRepository.deleteById(id);
            return ResponseEntity.noContent().build();
        }

        return ResponseEntity.notFound().build();
    }
}
```

```java
@RestController
@RequestMapping("/api/user")
public class UserController {

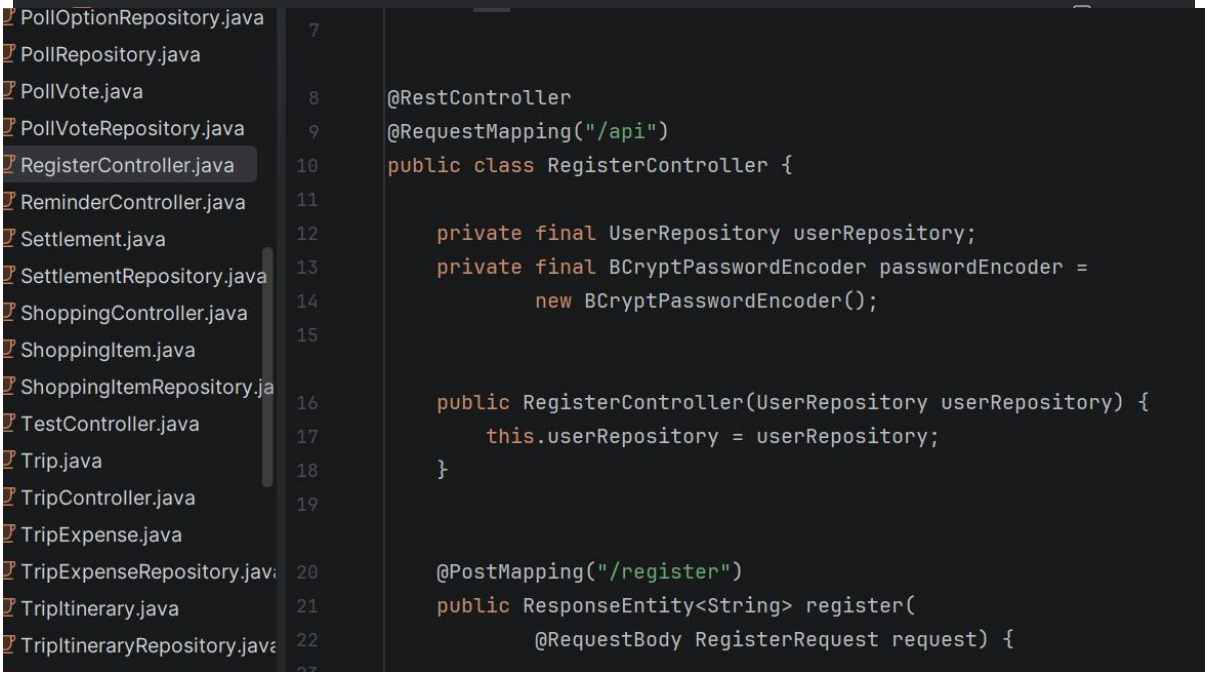
    private final UserRepository userRepository;

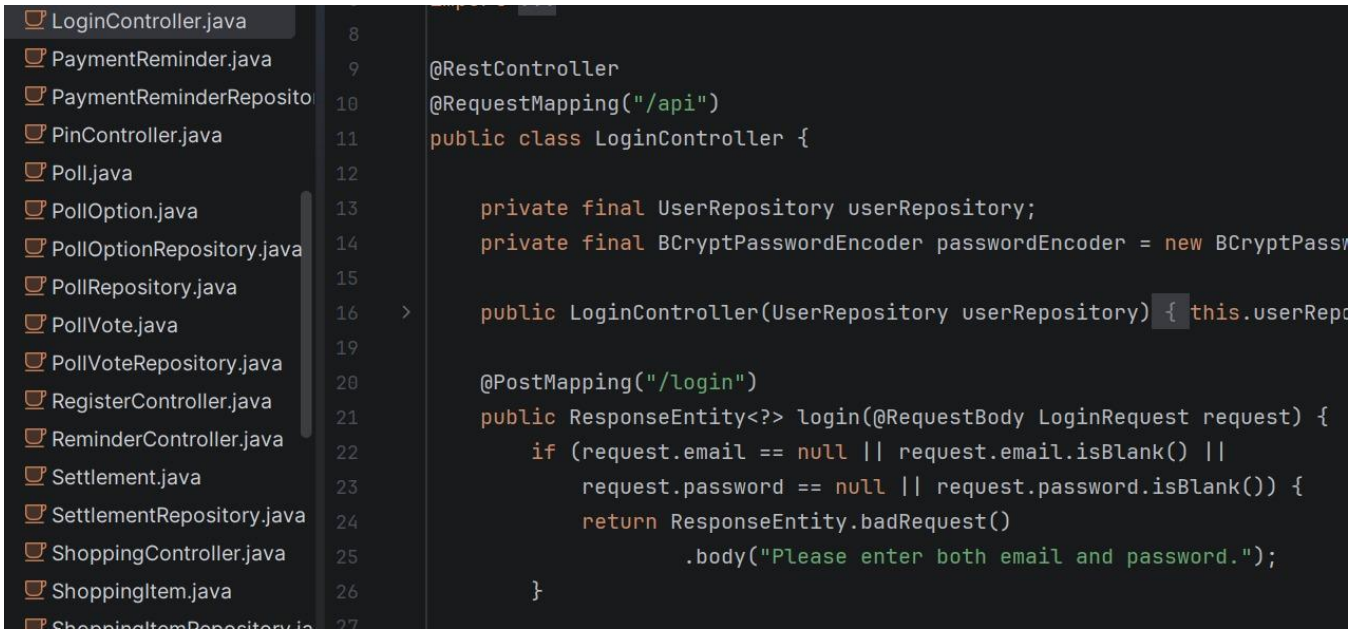
    public UserController(UserRepository userRepository) { this.userRepository = userRepository;

    @PostMapping("/upi")
    public ResponseEntity<?> updateUpiId(@RequestBody UpiUpdateRequest request) {
        if (request.userId == null || request.upiId == null || request.upiId.isBlank()) {
            return ResponseEntity.badRequest().body("User ID and UPI ID are required.");
        }

        Optional<User> userOpt = userRepository.findById(request.userId);
        if (userOpt.isEmpty()) {
            return ResponseEntity.status(HttpStatus.NOT_FOUND).body("User not found.");
        }

        User user = userOpt.get();
        user.setUpiId(request.upiId.trim());
        userRepository.save(user);
```





DATABASE ENTITY AND REPOSITORY:

```
package com.spliteasy.backend.model;

import jakarta.persistence.*;
import java.math.BigDecimal;
import java.time.LocalDate;

@Entity
@Table(name = "expenses")
public class Expense {
```

```
@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)
private Long id;

private String description;

private BigDecimal amount;

private String category;

private String paidBy;

private LocalDate expenseDate;

// Default constructor
public Expense() {
}

// Getters and Setters
public Long getId() {
   return id;
}

public void setId(Long id) {
   this.id = id;
}

public String getDescription() {
   return description;
}

public void setDescription(String description) {
   this.description = description;
}

public BigDecimal getAmount() {
   return amount;
}

public void setAmount(BigDecimal amount) {
   this.amount = amount;
}

public String getCategory() {
   return category;
}

public void setCategory(String category) {
   this.category = category;
}
```

```
    public String getPaidBy() {
        return paidBy;
    }

    public void setPaidBy(String paidBy) {
        this.paidBy = paidBy;
    }

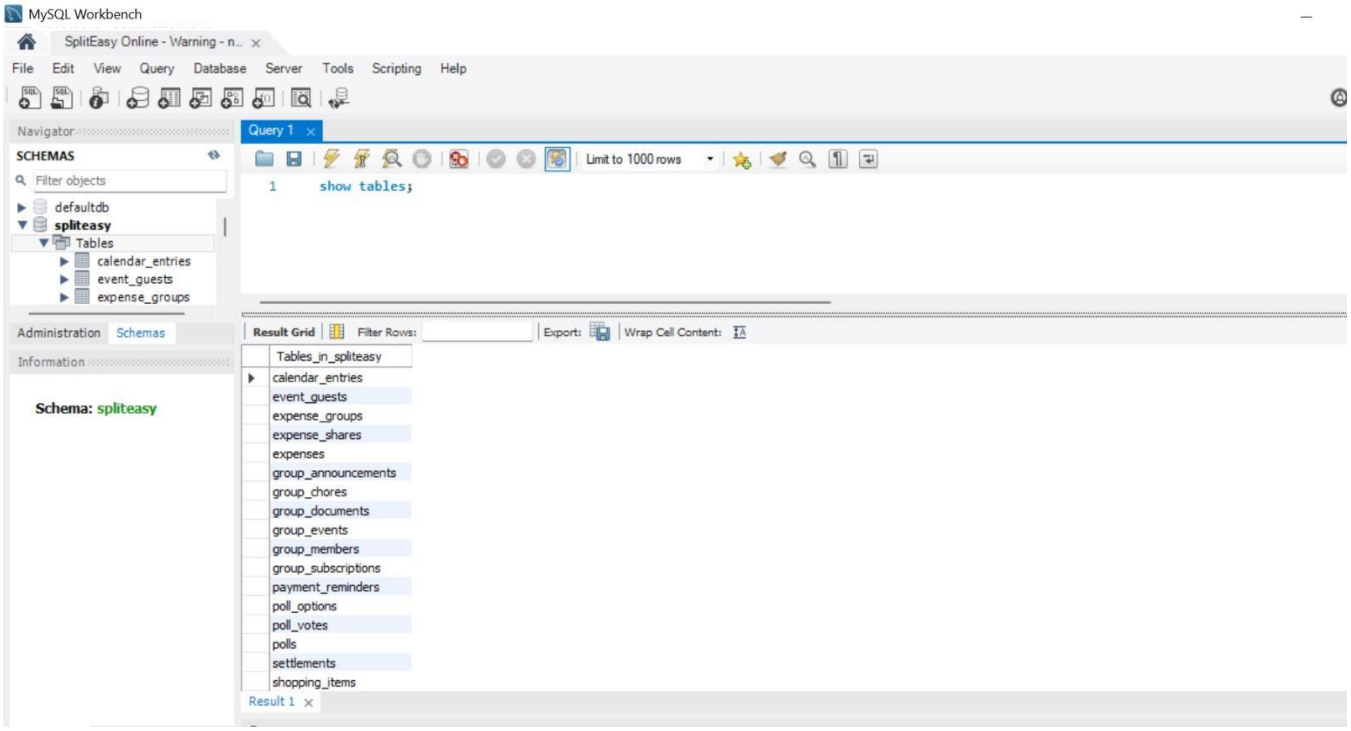
    public LocalDate getExpenseDate() {
        return expenseDate;
    }

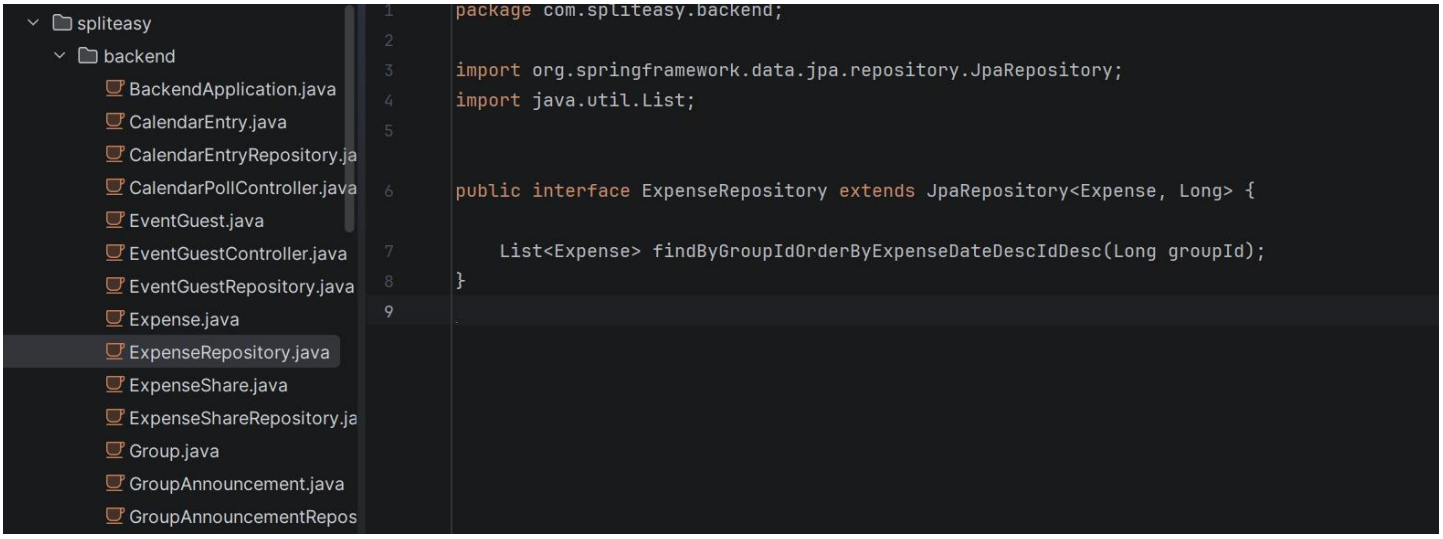
    public void setExpenseDate(LocalDate expenseDate) {
        this.expenseDate = expenseDate;
    }


}catch (Exception E){ E.printStackTrace();
                }

            } else if (e.getSource() == back) {
                System.exit(90);
            }
        }

        public static void main(String[] args) {
            new Login();
        }
    }
```




spliteasy
backend
BackendApplication.java
CalendarEntry.java
CalendarEntryRepository.ja
CalendarPollController.java
EventGuest.java
EventGuestController.java
EventGuestRepository.java
Expense.java
ExpenseRepository.java
ExpenseShare.java
ExpenseShareRepository.ja
Group.java
GroupAnnouncement.java
GroupAnnouncementRepos
package com.spliteasy.backend;
import org.springframework.data.jpa.repository.JpaRepository;
import java.util.List;
public interface ExpenseRepository extends JpaRepository<Expense, Long> {
List<Expense> findByGroupIdOrderByExpenseDateDescIdDesc(Long groupId);
}

- CONCLUSION

In conclusion, our SplitEasy expense-sharing management system, developed using Java, Spring Boot, REST APIs, and MySQL, provides an effective solution for managing and sharing expenses among groups. The application supports essential functionalities such as user registration and login, creating groups, adding members, recording shared expenses, calculating individual balances, and managing settlements. The backend and database work together to ensure the proper storage and retrieval of user, group, and expense information.

The project aims to simplify expense management by automating calculations and maintaining organized expense records. It helps reduce manual calculations, avoid confusion among group members, and provides better transparency in tracking shared expenses, outstanding balances, and repayments. Additional features, such as expense insights and group management facilities, further improve the overall expense-tracking experience.

Through this project, we gained valuable knowledge of Android application development using Java and XML, backend development using Spring Boot, REST API integration, and database management using MySQL. Overall, SplitEasy offers a simple and user-friendly approach to managing group expenses, making expense sharing more convenient, accurate, and organized.

- REFERENCES

1. Android Developers, Android Studio Documentation. Available at: https://developer.android.com/

2. Spring, Spring Boot Reference Documentation. Available at: https://docs.spring.io/spring-boot/

3. MySQL, MySQL Reference Manual. Available at: https://dev.mysql.com/doc/

4. GitHub, GitHub Documentation. Available at: https://docs.github.com/

5. Render, Render Documentation. Available at: https://render.com/docs

6. Aiven, Aiven for MySQL Documentation. Available at: https://aiven.io/docs/products/mysql

YOUTUBE (LINK BELOW)

https://www.youtube.com/watch?v=dqxeHPgNsRQ&list=PLQTYNpk8jwk0rYAEmI5uVcT1b9hE4M8T6